

MARS Summer Research Project

Explainable Feature Extraction
for Image Classification



Explainable



Efficient



Evolvable



Generalizable

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Project Overview

Aim



Develop a unified benchmarking framework to compare explainable feature extraction methods across multiple image datasets.

Objectives



Compare Handcrafted, GP and EOH feature generation methods



Evaluate classification performance



Measure computational efficiency

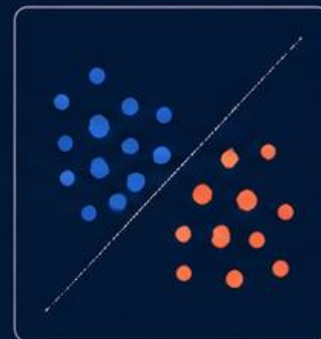
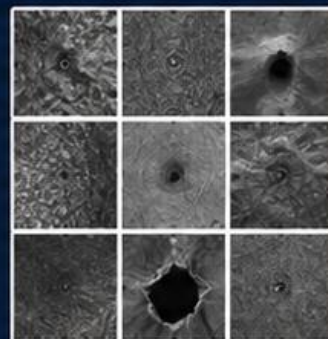


Assess suitability for real-time deployment



The project focuses on building reusable, **interpretable** and **efficient** feature extraction techniques that **generalize** across different image domains and application scenarios.

Explainable Features Bridging Images and Machine Learning



Explainable
Feature Extraction

Machine Learning
(Classification)

Progress Since Last Meeting

Major milestones achieved – remaining focus is on validation and final evaluation

Project Stage		Status
	FEI Benchmark Benchmark pipeline implemented and evaluation completed	 Complete
	KSDD2 Benchmark Official train/test protocol implemented and evaluation completed	 Complete
	Unified Evaluation Pipeline Common preprocessing, SVM classifier and metrics across all methods	 Complete
	Timing Framework Inference time measurement and throughput analysis implemented	 Complete
	Cross-validation Evaluate robustness across multiple folds	 In Progress
	Final Test Evaluation Locked test set evaluation on KSDD2 and final comparison across datasets	 In Progress



Where we are now

The core benchmarking framework is complete and has been applied to two distinct datasets. We now have comparative results for six feature extraction methods, including both performance and efficiency metrics.



Two benchmarks completed

FEI (facial expressions) and KSDD2 (industrial defect detection)



Six methods compared

Handcrafted, Restricted GP (Original & Modified), EOH, Original GP, Modified GP



Performance and efficiency measured

Accuracy metrics and inference time captured under a unified framework



The benchmarking framework is now largely complete.

The remaining work focuses on strengthening and finalising the experimental results.

FEI Benchmark Notebook

Facial Expression Images (FEI) Dataset



Purpose

Establish the benchmarking framework on a controlled dataset of facial expressions to compare explainable feature extraction methods.



Task

Binary classification: Neutral (0) vs Smiling (1)



Pre-processing

Grayscale conversion and resize to 64×64 pixels, intensity normalization to [0, 1]



Evaluation Protocol

Stratified split: 70% training, 15% validation, 15% test using a fixed random seed



Pipeline

Feature extraction → StandardScaler → Linear SVM → Metrics (Accuracy, Balanced Accuracy, Precision, Recall, F1)



Methods Evaluated

Handcrafted Features, Original GP, Modified GP, EOH (8-feature representations)

Sample Images

Neutral (0)

Smiling (1)

Dataset Statistics

 Total Images 700 images	 Neutral (0) 350 images	 Smiling (1) 350 images	 Subjects 50 individuals
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 14 images per subject (7 neutral, 7 smiling)



Outcome: Validated the benchmarking pipeline and demonstrated that evolved explainable features (EOH) outperform traditional handcrafted features on the FEI dataset.

FEI Results Summary

Performance Comparison Across Feature Extraction Methods

Classification Performance (Validation Set)			
	Method	# Features	Validation Accuracy (%)
1	Modified GP — Unrestricted	164	100.00
2	Original GP — Unrestricted	768	100.00
3	EOH-8	8	92.86
4	Restricted Original GP-8	8	89.29
5	Restricted Modified GP-8	8	82.14
6	Handcrafted-8	8	71.43

i All methods were evaluated using a Linear SVM classifier on the validation set.

Methods Evaluated		
1	Modified GP — Unrestricted	Genetic Programming with modified function set and no restrictions on depth/size
2	Original GP — Unrestricted	Genetic Programming with original function set and no restrictions on depth/size
3	EOH-8	Evolutionary Optimization of Handcrafted features (8-feature representation)
4	Restricted Original GP-8	Original GP with depth/size restrictions (8-feature representation)
5	Restricted Modified GP-8	Modified GP with depth/size restrictions (8-feature representation)
6	Handcrafted-8	Baseline handcrafted features (8-feature representation)



Both unrestricted GP variants achieved **100% validation accuracy** on the FEI dataset.



EOH-8 was the **strongest performing method** under the common 8-feature constraint.



Restricting feature dimensionality enables a **fair comparison** between explainable methods.



The **8-feature methods** remain computationally efficient while delivering competitive results.



Modified GP (Unrestricted) and Original GP (Unrestricted) achieve the highest performance on the FEI benchmark.
EOH-8 provides the best trade-off between performance and efficiency among 8-feature methods.



Takeaway: The FEI results confirm that evolved, explainable features—particularly those generated by GP and EOH—can outperform traditional handcrafted features while remaining interpretable and computationally efficient.

Why a Second Dataset? – KSDD2

Moving from a controlled facial dataset to a real-world industrial inspection benchmark

Motivation for KSDD2



Industrial Relevance

KSDD2 contains images of real industrial components with surface defects, reflecting a practical inspection scenario.



Different Domain

Significantly different from facial expressions in terms of visual content, texture, and class characteristics.



Official Train/Test Split

Provides a fixed, standardised split for fair and reproducible evaluation and comparison with prior work.



Better Assessment of Generalisation

Testing on a new domain helps assess how well each method generalises beyond the original dataset.



Stronger Conclusions

Results across two very different datasets give more robust insights into the strengths and trade-offs of each method.



KSDD2 Dataset Overview

Total Images
8,874

Categories
4

Training
~90%

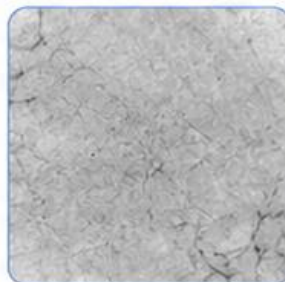
Test
~10%

Image Size
299 × 299

Sample Images from KSDD2

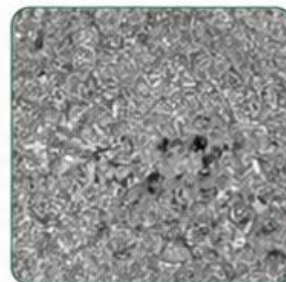
Crazing

(2,397 images)



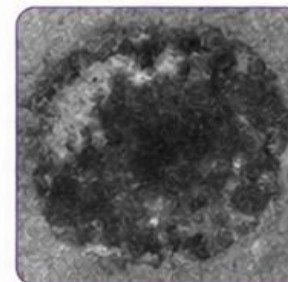
Inclusion

(2,351 images)



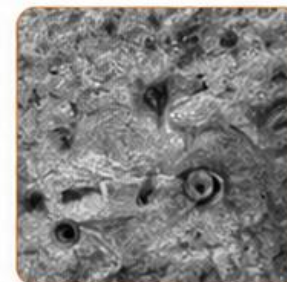
Patches

(2,060 images)



Pitted Surface

(2,066 images)



Why this matters:

By evaluating on KSDD2, we can determine which feature extraction methods are not only accurate, but also **robust** and **efficient** on **real-world** industrial inspection data.



Takeaway: KSDD2 provides a complementary and more challenging benchmark, allowing us to validate the generalisability and **practical applicability** of the explainable feature extraction methods developed and evaluated on FEI.

KSDD2 Benchmark Notebook

Unified evaluation pipeline for fair comparison across six feature extraction methods



Dataset Validation
Verified KSDD2 integrity and structure.
Confirmed official categories and train/test split.



Development Split
From the training portion, created stratified splits:
70% training, 15% validation, 15% development test.



Feature Extraction
Extracted features using six methods:
Handcrafted, Restricted GP (Original & Modified),
EOH, Original GP, Modified GP.



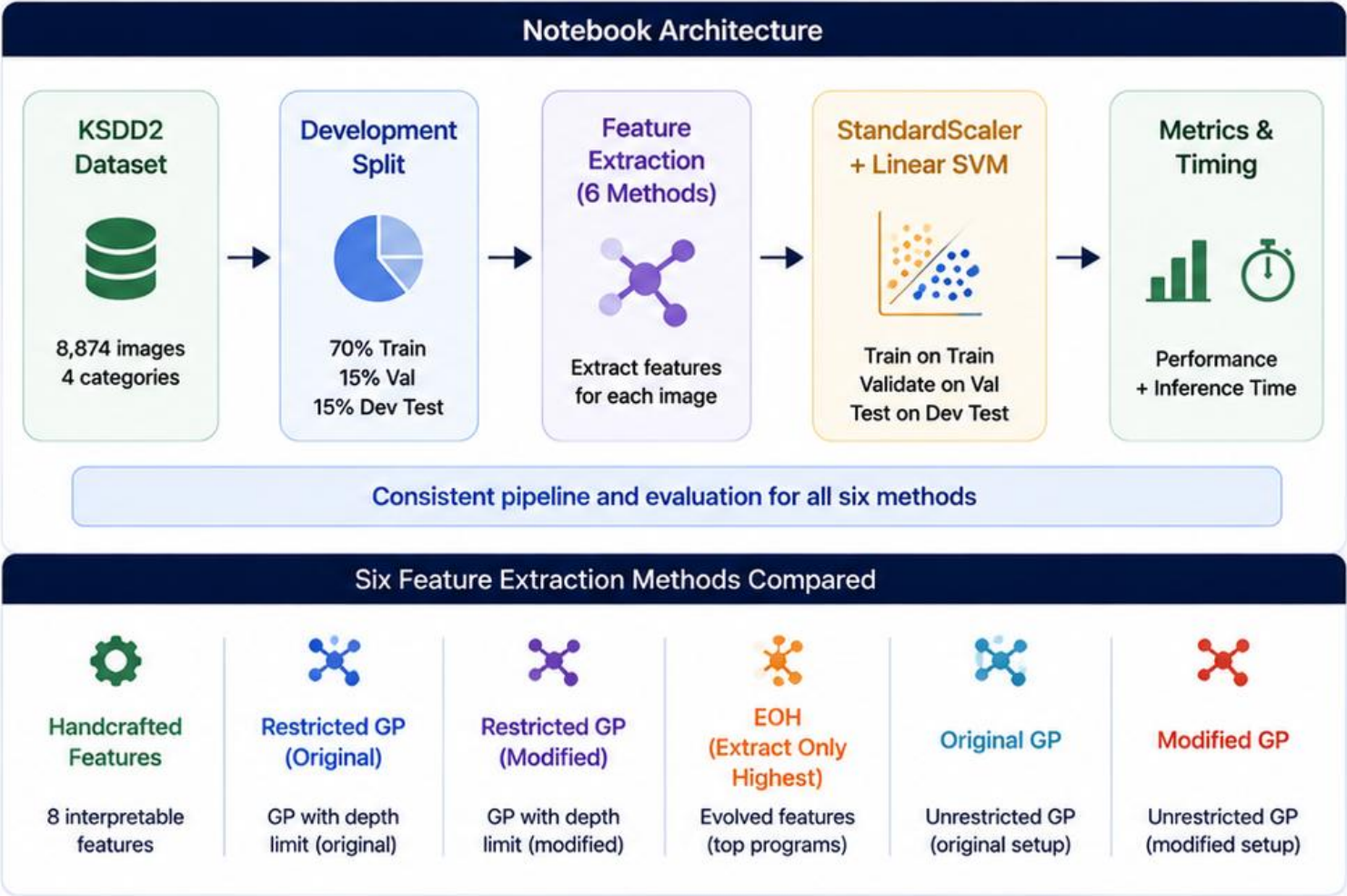
Common SVM Pipeline
Applied StandardScaler and Linear SVM for all methods
under identical settings to ensure fair comparison.



Performance Metrics
Computed Accuracy, Balanced Accuracy, Precision,
Recall, and F1 Score on the development test set.



Timing Benchmark
Measured end-to-end inference time per image
on the test set to assess computational efficiency.



Key Takeaway: The KSDD2 notebook implements a unified, reproducible evaluation framework, enabling rigorous comparison of **predictive performance** and **computational efficiency** across diverse, explainable feature extraction methods.

KSDD2 Classification Results

Performance Comparison on the Development Test Set

Performance Metrics (Development Test Set)

Method (# Features)	Balanced Accuracy (%)	Macro F1 (%)	Defect Recall (%)	Inference ms/image	95th Percentile ms	Estimated FPS
 Restricted Original GP-8 (8)	88.37	82.18	83.67	2.460	4.116	406.46
 EOH-8 (8)	86.99	80.54	81.63	0.789	2.052	1268.11
 Original GP — Unrestricted (153)	85.31	76.19	81.63	3.325	5.710	300.77
 Handcrafted-8 (8)	83.39	79.36	73.47	0.462	1.153	2165.24
 Restricted Modified GP-8 (8)	80.33	73.47	71.43	7.260	13.637	137.75
 Modified GP — Unrestricted (334)	79.97	72.65	71.43	4.737	6.348	211.12

Key Takeaways



Restricted GP (Original) achieves the highest Balanced Accuracy (**88.37%**), outperforming all other methods.



EOH-8 delivers a strong trade-off, achieving **86.99%** Balanced Accuracy just **1.38 percentage points** behind the best, while being **3.1× faster**.



Original GP (Unrestricted) ranks third (**85.31%** Balanced Accuracy), but does not surpass the restricted 8-feature GP (Original).



Handcrafted-8 is the fastest overall method (**2165.24 FPS**) but has lower predictive performance.



Restricted GP (Original) provides the best predictive performance, while EOH-8 offers an excellent balance of accuracy and efficiency.



Overall Conclusion: GP-based methods, particularly the original restricted variant, deliver the best predictive performance on the KSDD2 development test set. EOH-8 provides a highly competitive and efficient alternative.

KSDD2 Timing Statistics

Development Test Set

Method	Features (Dimensions)	Inference (ms/image)	95th Percentile (ms)	Estimated FPS
Restricted Original GP-8	8	2.460263	4.11612	406.460575
EOH-8	8	0.788576	2.05172	1268.108563
Original GP — Unrestricted	153	3.324772	5.71032	300.772478
Handcrafted-8	8	0.461842	1.15342	2165.244004
Restricted Modified GP-8	8	7.259738	13.63723	137.746014
Modified GP — Unrestricted	334	4.736664	6.34767	211.119063

Final Project Roadmap

Plan for the Final Week

Stage		Objective	Status
✓	Benchmark Framework	FEI and KSDD2 benchmarking notebooks completed	Complete
✓	Method Comparison	Handcrafted, GP, Modified GP and EOH evaluated	Complete
✓	Unified Evaluation Pipeline	Single pipeline for accuracy and efficiency metrics	Complete
✓	Timing Framework	End-to-end timing measurements implemented	Complete
↻	Cross-Validation	Verify robustness across multiple data splits	In Progress
📅	Official Test Evaluation	Run final evaluation on the locked KSDD2 test set	Planned
📊	Statistical Analysis	Summarise performance differences and significance	Planned
📄	Final Research Report	Consolidate results and prepare supervisor feedback	Planned



Expected Deliverables

- ✓ Validated benchmarking framework
 - FEI benchmark completed
 - KSDD2 benchmark completed
 - Timing analysis completed
- ↻ Cross-validation experiments
- 📅 Official KSDD2 test evaluation
- 📊 Final research conclusions



Timeline (Final Week)

- Mon–Tue:
Cross-validation across multiple splits
- Wed:
Official KSDD2 test evaluation
- Thu:
Statistical analysis and result interpretation
- Fri:
Final report and results summary



Project End Goal

Deliver a **robust and reproducible** benchmarking framework for explainable feature extraction, together with a **clear comparison** of Handcrafted Features, **Genetic Programming** and EOH across two benchmark datasets.



Overall: The benchmarking framework is now largely complete. The remaining work focuses on validating and finalising the experimental results to ensure robustness and reliability.